SITECORE® SYMPOSIUM 2021

Render Here, Render There, Render Everywhere

An introduction to rendering options for the web

Thomas Desmond

Developer Advocate @ Sitecore

October 6, 2021

Today's Speaker

Thomas Desmond

Developer Advocate @ Sitecore

- Frontend Developer Advocate
- 7+ years of development experience
- Lives in San Diego, California

@ThomasJDesmond

www.thetombomb.com

Agenda

01 Important Performance Indicators

02 What is Rendering on the Web?

03 Rendering Options vs. Performance Indicators

04 Sitecore Offerings

05 No One Size Fits All

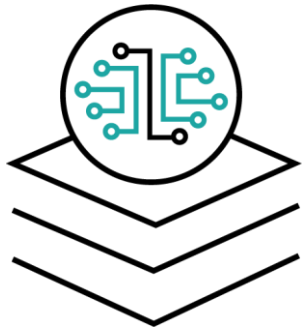
© 2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

Performance Indicators – The Beginning

Time To First Byte (TTFB)

What: The time between user requesting page to the browser getting the first byte of information from the server.

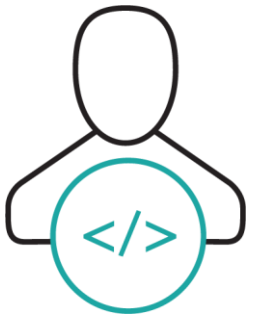
Why: Slow TTFB means slow load times. It's important how fast we can get data to the browser.



First Contentful Paint (FCP)

What: The time for the first thing to be painted to the screen.

Why: A blank page is discouraging to a visitor. Seeing something loaded is better than nothing.



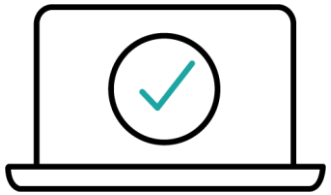
© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

Performance Indicators – The End

Largest Contentful Paint (LCP)

What: The time it takes for the largest image or text block to become visible.

Why: The largest item is usually the important piece of a page. This is what the visitor came for.



Time To Interactive (TTI)

What: The time to become FULLY interactive

Why: A page may look like it is all their visually but feels frozen when user tries to interact

© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

Web App Rendering

- What is Rendering?
- The Big Three Options
  - Server Side Rendering
  - Client Side Rendering
  - Static Rendering

© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

Rendering Options – Sever Side Rendering

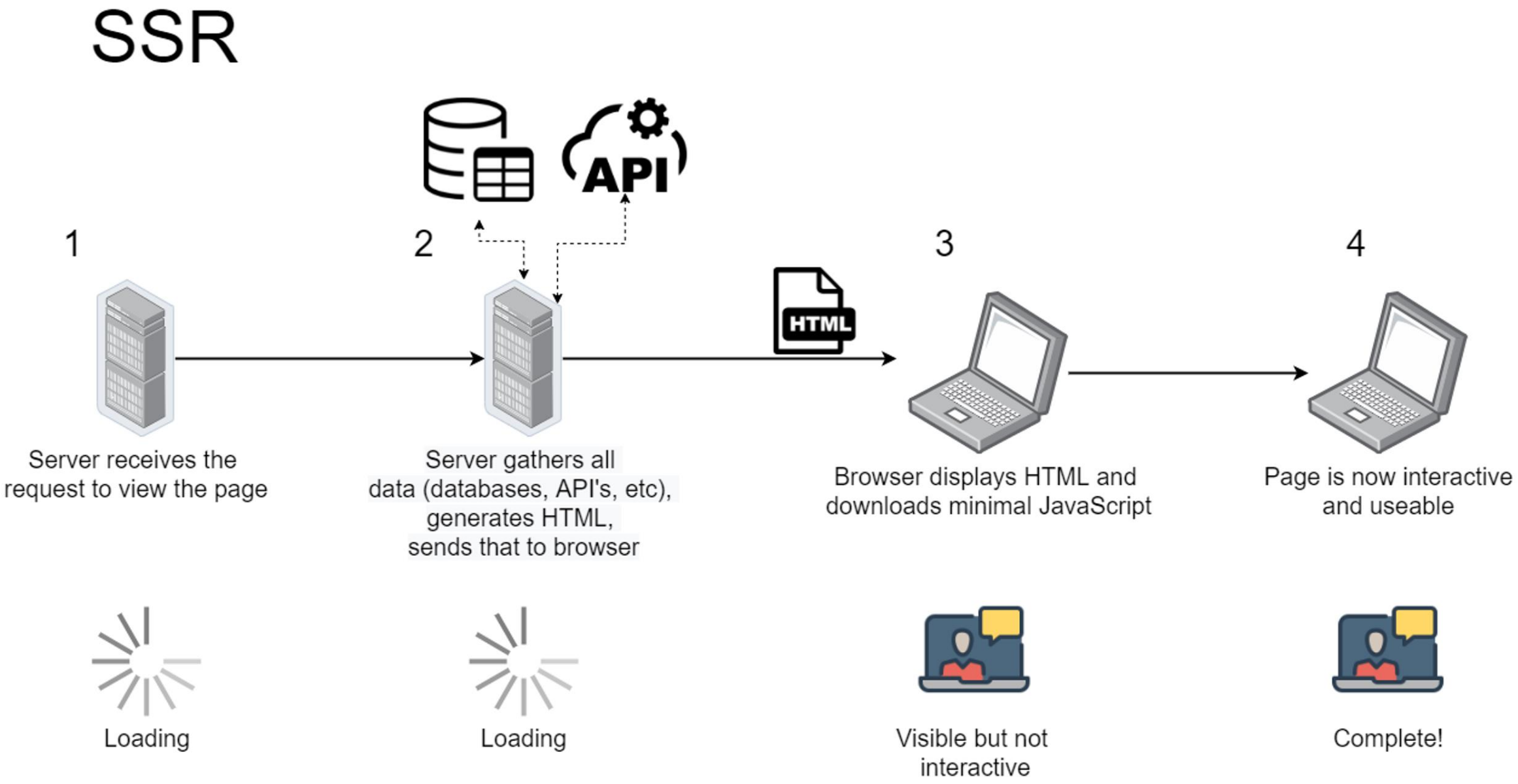
Server Side Rendering (SSR)

- Putting the work of rendering the HTML on the server
- Server more powerful than visitor's machine
- Browser executes minimal JavaScript

- **The Good:** 👍 FCP, TTI, SEO
- **The Bad:** 👎 TTFB, Blank Page

© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

Server Side Rendering Flow



Rendering Options – Client Side Rendering

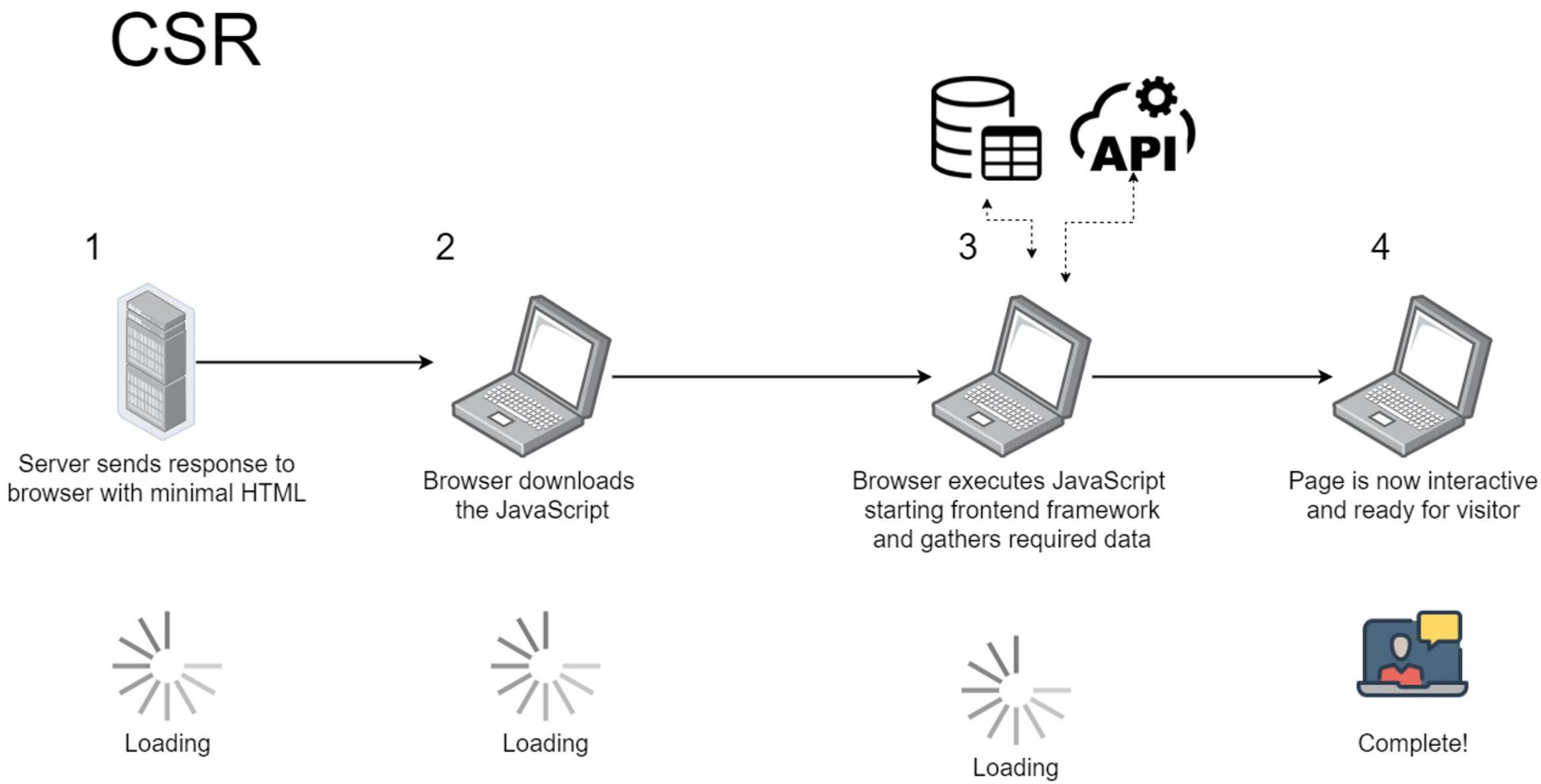
Client Side Rendering (CSR)

- Server sends minimal HTML with links to JavaScript to download
- Site visitor's browser does the rendering work

- **The Good:** 👍 TTFB, Flexibility, Low server costs
- **The Bad:** 👎 SEO, TTI, JavaScript performance

© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

Client Side Rendering Flow



Rendering Options – Static Site Generation

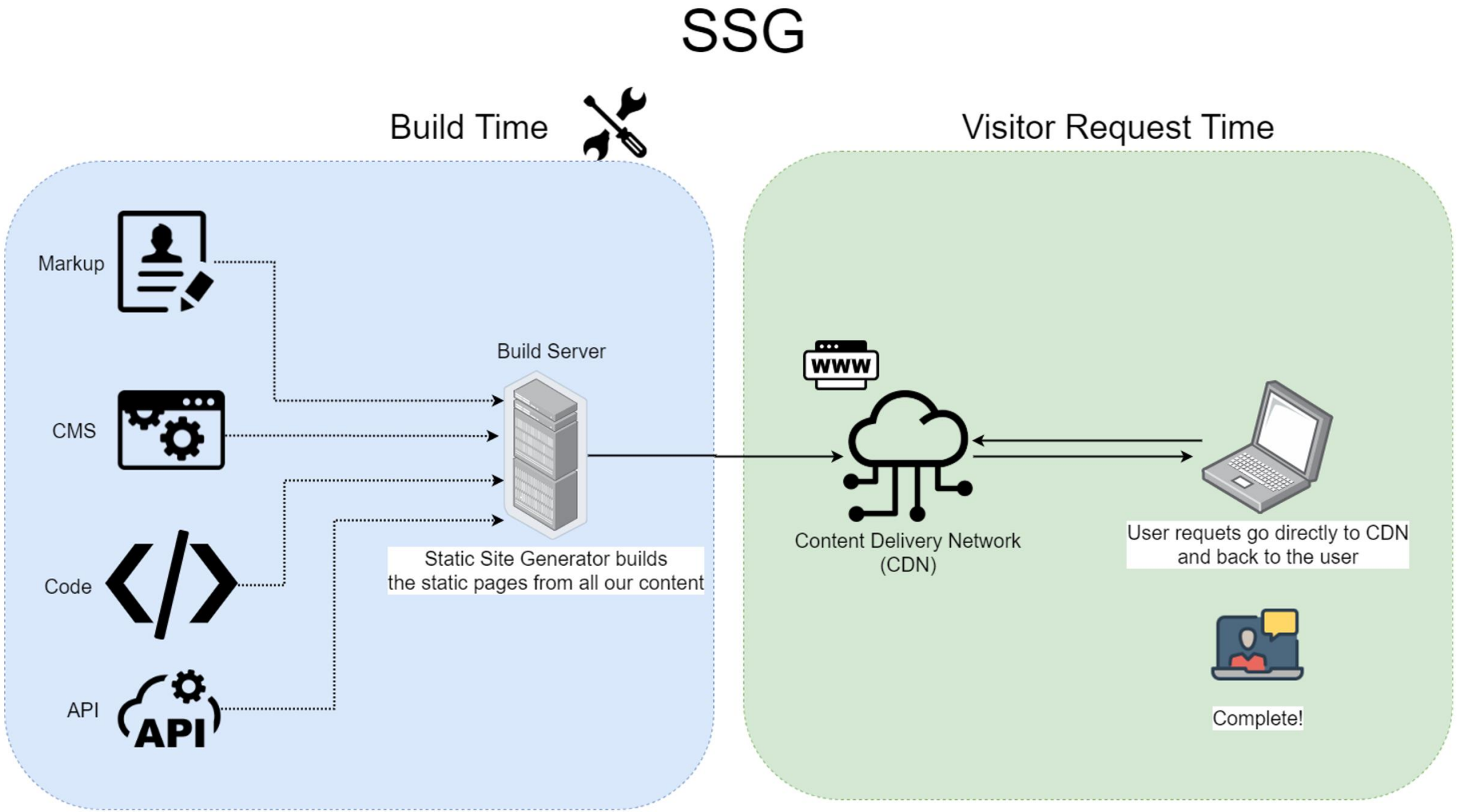
Static Site Generation (SSG)

- Ahead of Time Rendering
- Build out static pages before anyone even visits the site
- Able to be hosted on low-cost Content Delivery Network (CDN)

- **The Good:** 👍 TTFB, FCP, TTI, SEO
- **The Bad:** 👎 Not flexible, Rehydration often required

© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

Static Site Generation Flow

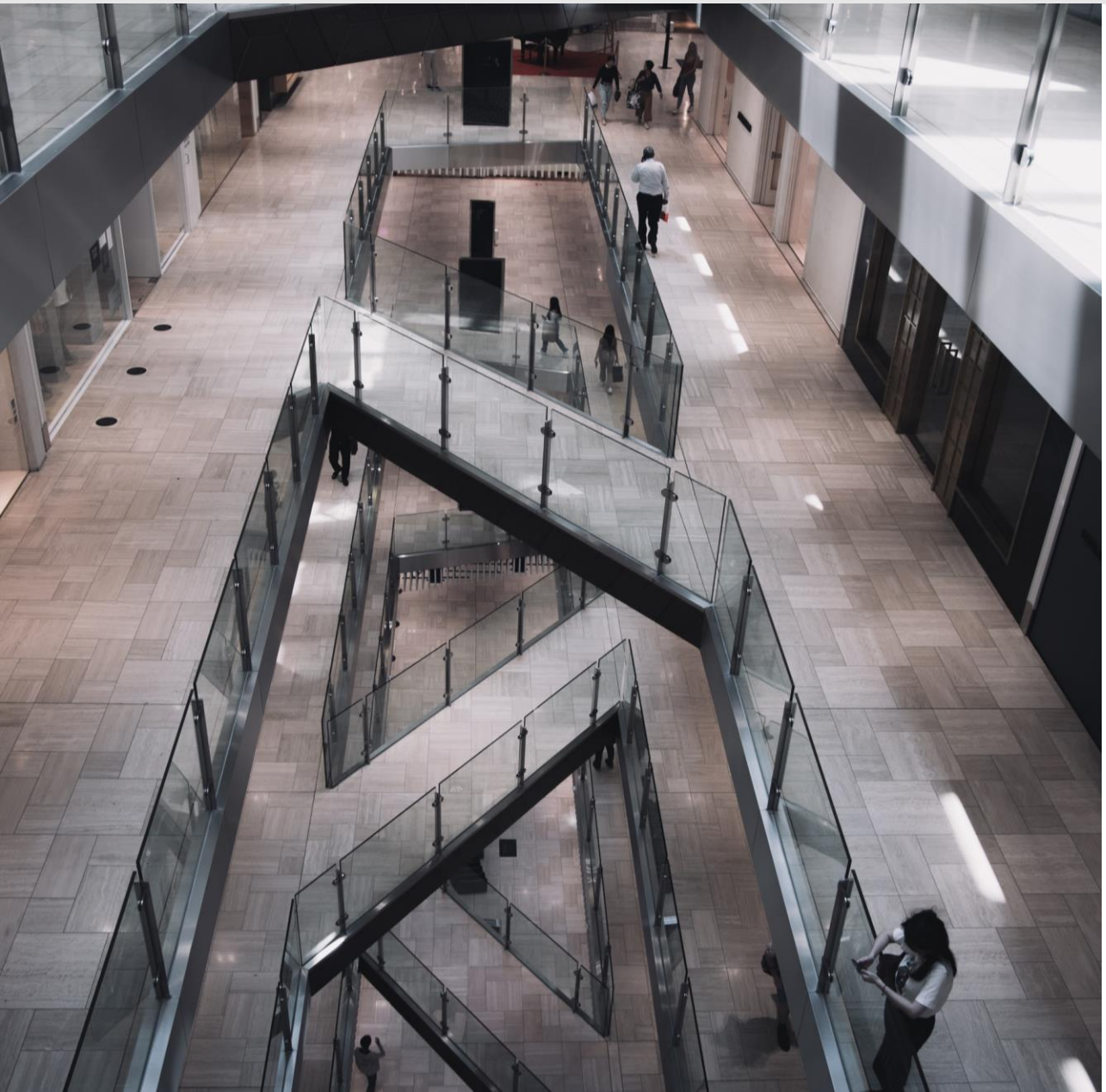


Putting Them All Together

SSR	CSR	SSG
Rendering work completed on server	Rendering done on the user's machine within browser	Rendering completed at build time before users visits site
✔ Personalization	✔ Personalization	✕ Personalization requires rehydration
👍 SEO, FCP, TTI 👎 TTFB, Blank Page Syndrome	👍 FCP, TTFB, Low server costs 👎 SEO, TTI, Requires JavaScript	👍 SEO, TTFB, FCP, TTI 👎 Inflexible, build times
Common Frameworks: ASP.NET, Next.js, PHP (Laravel), Node.js	Common Frameworks: React, Angular, Vue	Common Frameworks: Next.js, Gatsby, Hugu, Nuxt

© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

Advanced Rendering Options



- Why limit yourself to only one?
- Advanced Options
  - Two Pass Rendering with Hydration
  - SSR with CSR
  - SSG with CSR

© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

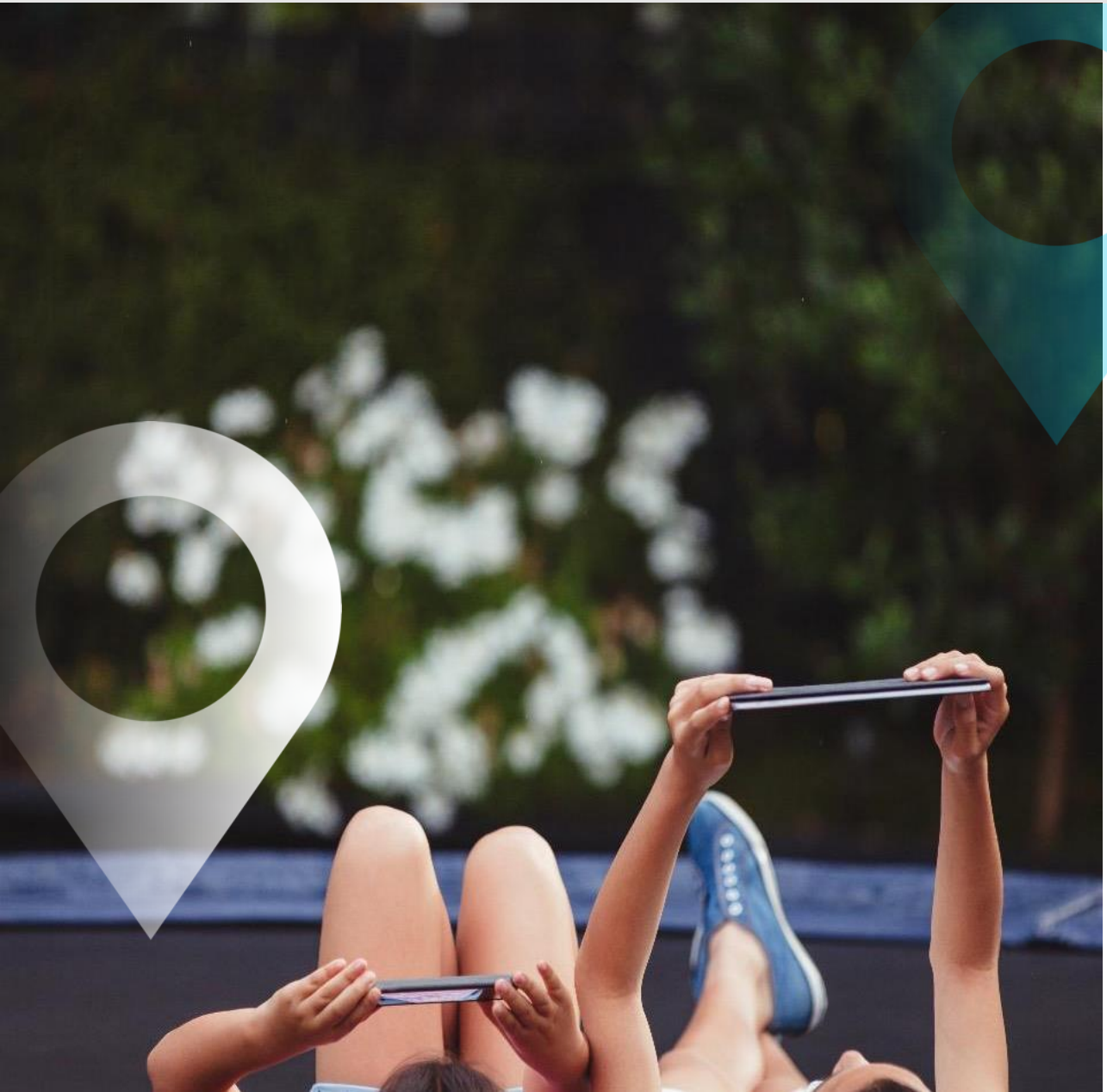
Sitecore Product Offerings for SSR, CSR, SSG

SITECORE®

SSR	Sitecore MVC SXA Sitecore Headless Services (.NET, Next.js, Angular, React, Vue)
CSR	Sitecore Headless Services (Angular, React, Vue)
SSG	Sitecore Headless Services (Next.js) Experience Edge for XM and Content Hub

© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

No one size fits all



- Considerations
  - Performance indicators
  - SEO
  - Update frequency
  - Visitor location
  - Team development skills
  - Costs
  - Long term
  - And More
- Rule of least power

© 2001-2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

Agenda

01 Important Performance Indicators

02 What is Rendering on the Web?

03 Rendering Options vs. Performance Indicators

04 Sitecore Offerings

05 No One Size Fits All

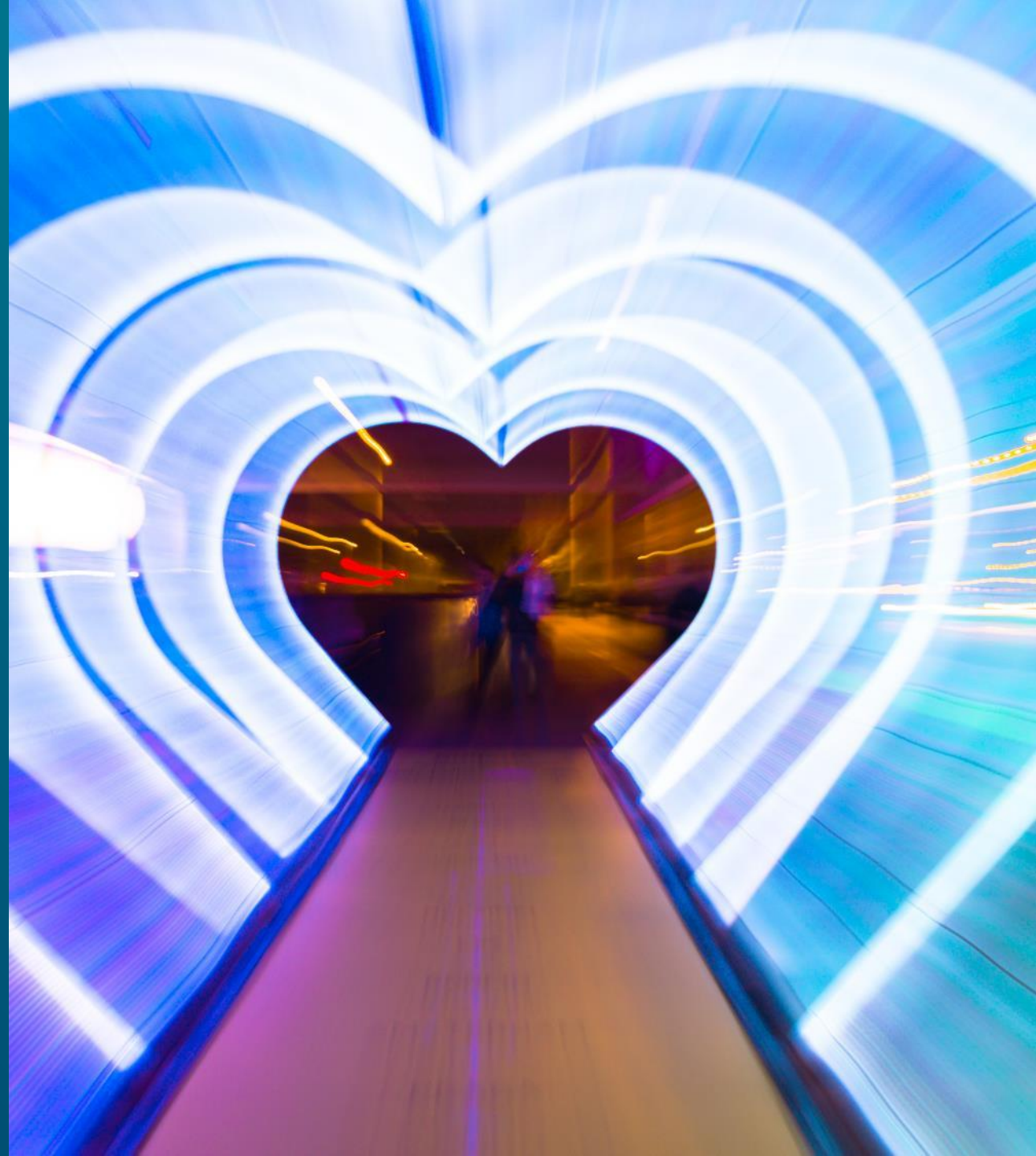
© 2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

SITECORE®
SYMPOSIUM
2021

UNFORGETTABLE
DIGITAL
CONNECTIONS

Thank you

FOR DISCUSSION PURPOSES ONLY.

Sitecore Confidential and Proprietary. ©2021 Sitecore Corporation A/S. Sitecore® and Own the Experience® are registered trademarks of Sitecore Corporation A/S. All other brand names are the property of their respective owners.

#SitecoreSYM